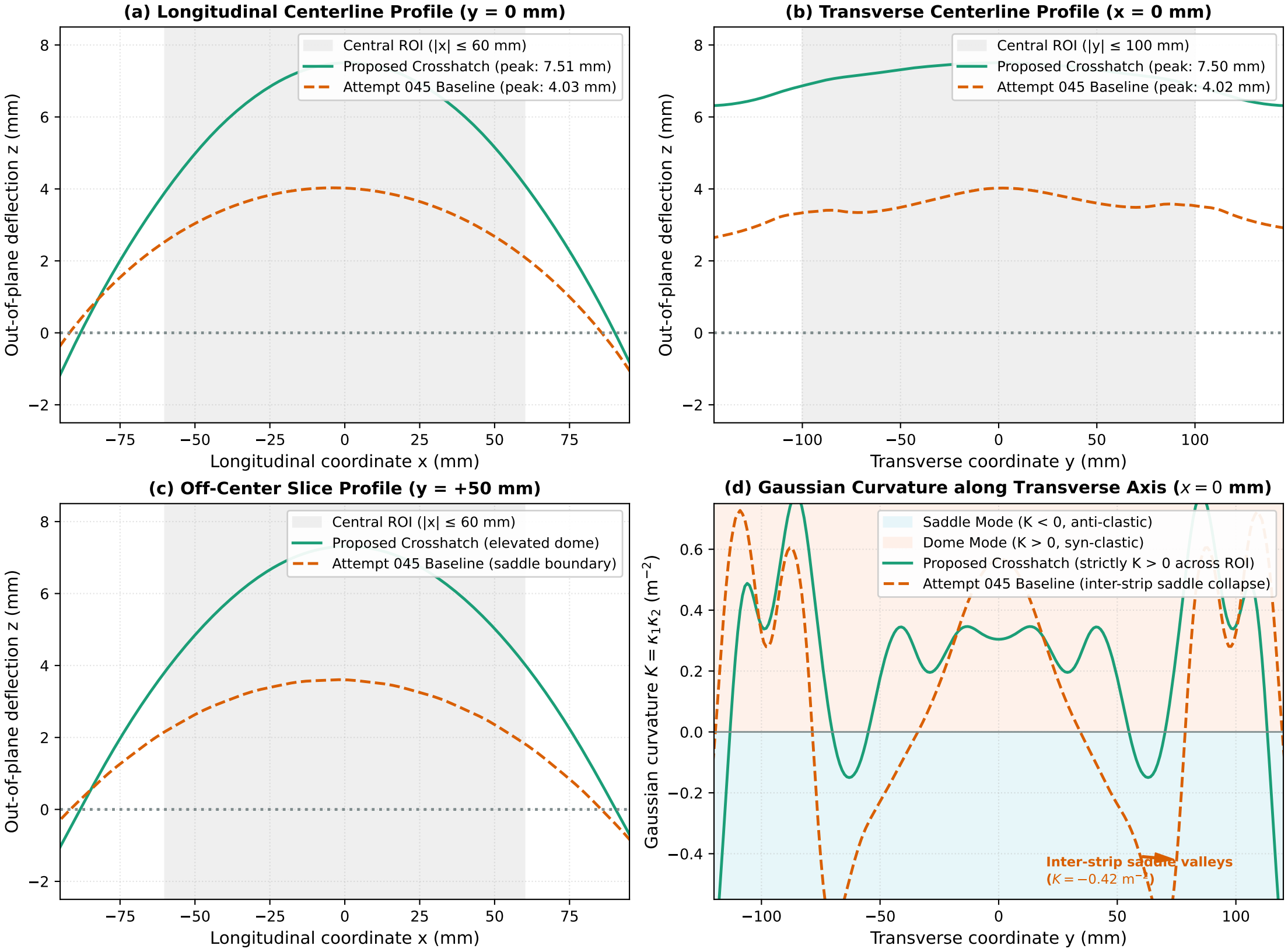
(a) Longitudinal Centerline Profile (y = 0 mm)
Central ROI (|x| ≤ 60 mm)
Proposed Crosshatch (peak: 7.51 mm)
Attempt 045 Baseline (peak: 4.03 mm)
Out-of-plane deflection z (mm)
Longitudinal coordinate x (mm)
(b) Transverse Centerline Profile (x = 0 mm)
Central ROI (|y| ≤ 100 mm)
Proposed Crosshatch (peak: 7.50 mm)
Attempt 045 Baseline (peak: 4.02 mm)
Out-of-plane deflection z (mm)
Transverse coordinate y (mm)
(c) Off-Center Slice Profile (y = +50 mm)
Central ROI (|x| ≤ 60 mm)
Proposed Crosshatch (elevated dome)
Attempt 045 Baseline (saddle boundary)
Out-of-plane deflection z (mm)
Longitudinal coordinate x (mm)
(d) Gaussian Curvature along Transverse Axis (x = 0 mm)
Saddle Mode (K < 0, anti-clastic)
Dome Mode (K > 0, syn-clastic)
Proposed Crosshatch (strictly K > 0 across ROI)
Attempt 045 Baseline (inter-strip saddle collapse)
Inter-strip saddle valleys
(K = −0.42 m⁻²)
Gaussian curvature K = κ1κ2 (m⁻²)
Transverse coordinate y (mm)